

2nd International Workshop on Rapid Continuous Software Engineering (RCoSE 2015)

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Abstract—Continuous software engineering refers to the organizational capability to develop, release and learn from software in very short rapid cycles, typically hours, days or a very small numbers of weeks. This requires not only agile processes in teams but in the complete research and development organization. Additionally, the technology used in the different development phases, like requirements engineering and system integration, must support the quick development cycles. Finally, automatic live experimentation for different system alternatives enables fast gathering of required data for decision making. The workshop, the second in the series after the first one at ICSE 2014, aims to bring the research communities of the aforementioned areas together to exchange challenges, ideas, and solutions to bring software engineering a step further to being a holistic continuous process. The workshop program is based on eight papers selected in the peer-review process and supplemented by interaction and discussions at the workshop. The topics range from agile methods, continuous software engineering practices to specific techniques, like visualization and testing.

I. INTRODUCTION

Today, software development is conducted in increasingly turbulent business environments. Typically, fast-changing and unpredictable markets, complex and changing customer requirements, pressures of shorter time-to-market, and rapidly advancing information technologies are characteristics found in most software development projects. To address this situation, agile practices [1] advocating flexibility, efficiency and speed are seen as increasingly attractive by software development companies.

While many software development companies have indeed succeeded in adopting agile practices in parts of their organization, the focus for many organisations is predominantly at the team level. The other functions in the organization of many companies, including customer relations, product management, R&D management and software release, still work in traditional slow cycles, measured in months and years. However, the first examples of companies, for instance in the Web 2.0 industry, now start to appear that operate their entire R&D cycle in a rapid continuous fashion. For instance, Amazon releases new software every 11 seconds and carefully measures the value that this software provides to customers [2], [3].

Rapid continuous software engineering refers to the organizational capability to develop, release and learn from software in fast parallel cycles, typically hours, days or a very small numbers of weeks. This includes determining new functionality to build, prioritizing the most important functionality, evolving

and refactoring the architecture, developing the functionality, validating it, releasing it to customers and collecting feedback from the customers to inform the next cycle of development [4].

The capability to perform all these activities in days or a few weeks requires significant changes in the entire software engineering approach, including parallelizing activities, empowering cross functional teams to allow for rapid decision making and light weight coordination across teams. It also requires significant technical advances in the engineering infrastructure, including continuous integration and deployment, collection of post-deployment product usage data, support for running automatic live experiments to evaluate different system alternatives, e.g., A/B testing.

In summary, as reaching the goal of rapid continuous software engineering is a holistic endeavor, it cannot be addressed only by research in the area of process aspects in software engineering, specifically, agile software development processes. Instead it requires additionally to address at least the following three aspects:

- 1) The *technology* for all different phases of software engineering like requirements engineering, architecture and design, implementation, and validation and verification must be adapted to support for parallel engineering of software.
- 2) The whole research and development *organization* must adapt to be compatible with the agile process in the development teams.
- 3) Approaches for live experimentation must be available and the results must be appropriately fed back into the artifacts of the different phases which are affected by the results of the live experimentation. [5], [6]

Consequently, the workshop aims to bring the research communities of the aforementioned areas together to exchange challenges, ideas, and solutions to bring software engineering a step further to being a holistic and rapid continuous process.

The workshop is the continuation of the successful first workshop on Rapid Continuous Software Engineering at ICSE 2014 [7] which attracted many submissions and a large number of participants.

II. ACCEPTED PAPERS

The workshop is open to all ICSE 2015 attendees and includes presentations based on the peer-reviewed papers. The

topics of the papers and the discussion at the workshop cover agile development methods, the different phases of continuous software engineering [8], e.g., continuous deployment, continuous delivery, continuous experimentation as well as specific techniques applied in continuous software engineering, e.g., visualization, testing, API design.

Specifically, the following papers were accepted for presentation at the workshop (listed in alphabetical order of the first author):

- Imed Hammouda, Eric Knauss, Leonardo Costantini: Continuous API-Design for Software Ecosystems
- Sebastian Klepper, Stephan Krusche, Sebastian Peters, Bernd Bruegge, Lukas Alperowitz: Introducing Continuous Delivery of Mobile Apps in a Corporate Environment: A Case Study
- Eric Knauss, Mirosław Staron, Wilhelm Meding, Ola Söder, Agneta Nilsson, Magnus Castell: Supporting Continuous Integration by Code-Churn Based Test Selection
- Aapo Koski, Tommi Mikkonen: Rolling out a mission critical system in an agilish way
- Eero Laukkanen, Mika Mäntylä: Build Waiting Time in Continuous Integration – An Initial Interdisciplinary Literature Review
- Marko Leppänen, Terhi Kilamo, Tommi Mikkonen: Towards Post-Agile Development Practices Through Productized Development Infrastructure
- Anna-Liisa Mattila, Timo Lehtonen, Kari Systä, Tommi Mikkonen, Henri Terho: Mashing Up Software Issue Management, Development and Usage Data
- Olli Rissanen, Jürgen Münch: Continuous Experimentation in the B2B Domain: A Case Study

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REFERENCES

- [1] J. Highsmith and A. Cockburn, "Agile software development: The business of innovation," *IEEE Computer*, vol. 34, no. 9, pp. 120–122, 2001.
- [2] J. Jenkins, "Amazon's technical evolution," 2011, presentation at Intuit Tech Forum, February 2011.
- [3] W. Vogel, "Keynote at Amazon Re:Invent," 2012, <http://www.informationweek.com/d/d-id/1107599>.
- [4] H. H. Olsson, H. Allahyari, and J. Bosch, "Climbing the "stairway to heaven" - a multiple-case study exploring barriers in the transition from agile development towards continuous deployment of software," in *EUROMICRO-SEAA*, V. Cortellessa, H. Muccini, and O. Demirörs, Eds. IEEE Computer Society, 2012, pp. 392–399.
- [5] S. Thomke, *Experimentation Matters: Unlocking the Potential of New Technologies for Innovation*. Harvard Business Review Press, 2003.
- [6] R. Kohavi, T. Crook, and R. Longbotham, "Online experimentation at Microsoft," in *Proc. of the Third Workshop on Data Mining Case Studies and Practice Prize*, 2009.
- [7] M. Tichy, J. Bosch, M. Goedicke, and M. Larsson, Eds., *1st International Workshop on Rapid Continuous Software Engineering, RCoSE 2014, Hyderabad, India, June 3, 2014*. ACM, 2014.
- [8] B. Fitzgerald and K. Stol, "Continuous software engineering and beyond: trends and challenges," in *1st International Workshop on Rapid Continuous Software Engineering, RCoSE 2014, Hyderabad, India, June 3, 2014*, M. Tichy, J. Bosch, M. Goedicke, and M. Larsson, Eds. ACM, 2014, pp. 1–9. [Online]. Available: <http://doi.acm.org/10.1145/2593812.2593813>